

` C++ Mock Assessment-1st December 2025

Test Reference No: PLMAcademy-Nov2025-General-Batch

Total Marks : 30

Test Duration :

Instructions to Candidate

1. All questions are compulsory/mandatory.
2. This is a closed book test.
3. There is no negative marking.
4. Wherever there is ambiguity, make suitable assumptions and proceed.
Document the assumptions in the answer.
5. Assume C++ 11 language standard.
6. Assume usage of Microsoft Visual Studio development platform.

Question No.	Question	Marks
1	What will be the output of the following code? <pre>#include <iostream> using namespace std ; class X { public : X() { cout << "Default constructor called" << endl ; } X(int i) { cout << "Parameterized constructor called" << endl ; } }; int main () { X x1 = 10 ; return 0 ; }</pre>	1
2	Complete the below code by calling the operator-> function in the main() function and mention the output. <pre>#include <iostream></pre>	2

	<pre> using namespace std ; class X { public: void operator-> () { cout << "X::operator-> function called" << endl ; } }; int main () { X x1 ; // Write code here to call the operator-> function return 0 ; } </pre>	
3	What will be the output of the following code? <pre> #include <iostream> using namespace std ; int main () { try { throw 0 ; } catch(...) { cout << "Exception caused" << endl ; } return 0 ; } </pre>	1
4	What is stack unwinding? Explain with a suitable example.	2
5	Explain what is VTable and VPTR?	2
6	What is a function declaration? Explain with minimal code.	1
7	The following code works perfectly well: <pre> #include <iostream> using namespace std; int main () { f(10); f(20, 30); f(40, 10, 10); } </pre>	1

	<pre>return 0; }</pre> Conditions: f is a global function. f does not have variable argument list i.e. it does not have the signature containing ellipsis (...) f is not overloaded, there is just one function f. Give a suitable declaration for function f.	
8	What is the output of the following program? <pre>#include <iostream> using namespace std; int main() { int a[] = {1, 5, 3, 7, 5, 9, 7, -1, 9, 3 }; int i = 0; while(a[i] > 0) { cout << a[i] << endl; i = a[i+1]; } return 0; }</pre>	1
9	Explain Smart Pointers. Give declaration for unique pointer, shared pointer, weak pointer.	2
10	What will be the output of the following program? <pre>#include <iostream> using namespace std; int main() { int ai[4] = {10, 30, 40, 60}; int *pi = &ai[3]; cout << pi[-2] << endl; return 0; }</pre>	1

11	Write a function which counts the number of digits (i.e. characters from 0 to 9) in the string; <pre> int strCountDigits(char * pStr) { // Write code here } int main () { int nCount = strCountDigits("a3f4"); return 0; } </pre>	2
12	What is wrong in the following code? <pre> #include <iostream> using namespace std ; int main() { int *p = new int[3] ; int *q = p ; delete [] q ; return 0; } </pre>	2
13	Will the following program compile and execute. If so what will be the output of the following program? <pre> #include <iostream> using namespace std; template <typename T> void func(T a, T B) { cout << " In Function func" << endl; } int main() { func(1, 2.0); } </pre>	1
14	What will be the output of the following program? <pre> #include <iostream> using namespace std; namespace API { inline namespace v1 { </pre>	1

	<pre> void fetch() { cout << "v1" << endl; } } namespace v2 { void fetch(){cout <<"v2" << endl; } } int main() { API::fetch(); API::v2::fetch(); } </pre>	
15	Write a program that returns the elements in a vector that are smaller than their adjacent left and right neighbors. Input: Vector Elements. Ex : 1 2 5 0 3 1 7 Output: 0 1	3
16	Write a program to implement exception handling in a vector.	2
17	Complete the function signature for function f in following code: // No global variables are to be used. <pre> void f(_____) { (*g)(); } int k() { return 2 ; } int main() { f (&k); return 0 ; } </pre>	1
18	What will be the output of the following code? Will it give	1

	the incremented value of i in main(). If not , explain why? <pre>#include <iostream> using namespace std; void increment(int i) { i = i + 1; } int main() { int i = 10; increment(i); cout << "Incremented value of i is " << i << endl; return 0; }</pre>	
19	What is the output of the following code? <pre>#include <iostream> using namespace std ; enum Color_e { Red, Green, Blue } ; int main () { Color_e Blue = Color_e::Blue ; cout << Blue << endl ; return 0 ; }</pre>	1
20	Will the following code compile and link? If not, why not? If yes, what will be the behavior. <pre>class X { public: virtual void f () = 0 ; }; int main () { X x1 ; return 0 ; }</pre>	2

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